

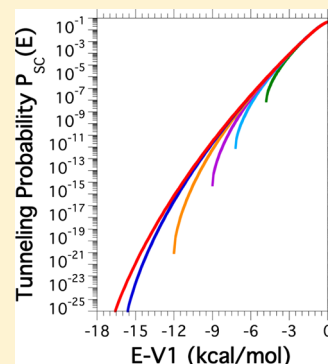
Improved Multidimensional Semiclassical Tunneling Theory

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S Supporting Information

ABSTRACT: We show that the analytic multidimensional semiclassical tunneling formula of Miller et al. [Miller, W. H.; Hernandez, R.; Handy, N. C.; Jayatilaka, D.; Willets, A. *Chem. Phys. Lett.* **1990**, *172*, 62] is qualitatively incorrect for deep tunneling at energies well below the top of the barrier. The origin of this deficiency is that the formula uses an effective barrier weakly related to the true energetics but correctly adjusted to reproduce the harmonic description and anharmonic corrections of the reaction path at the saddle point as determined by second order vibrational perturbation theory. We present an analytic improved semiclassical formula that correctly includes energetic information and allows a qualitatively correct representation of deep tunneling. This is done by constructing a three segment composite Eckart potential that is continuous everywhere in both value and derivative. This composite potential has an analytic barrier penetration integral from which the semiclassical action can be derived and then used to define the semiclassical tunneling probability. The middle segment of the composite potential by itself is superior to the original formula of Miller et al. because it incorporates the asymmetry of the reaction barrier produced by the known reaction exoergicity. Comparison of the semiclassical and exact quantum tunneling probability for the pure Eckart potential suggests a simple threshold multiplicative factor to the improved formula to account for quantum effects very near threshold not represented by semiclassical theory. The deep tunneling limitations of the original formula are echoed in semiclassical high-energy descriptions of bound vibrational states perpendicular to the reaction path at the saddle point. However, typically ab initio energetic information is not available to correct it. The Supporting Information contains a Fortran code, test input, and test output that implements the improved semiclassical tunneling formula.



1. INTRODUCTION

Quantum mechanical tunneling through reaction barriers has long been recognized as an important phenomenon for thermal kinetics processes, especially at lower temperatures. While tunneling can only be exactly represented by rigorous solution of the Schrodinger equation, many approximate approaches to tunneling exist and are incorporated into major theoretical kinetics software packages such as Polyrate¹ and Multiwell.² Tunneling for thermal processes necessarily weighs most heavily tunneling near the top of the reaction barrier. However, there are a variety of experiments and processes that are sensitive to deep tunneling well below the top of the barrier. For example, photodetachment experiments³ on anions can place the neutral molecule deep within a well whose exit via tunneling can be detected. Similarly, tunneling-controlled synthesis⁴ creates products from initially synthesized long-lived metastable species whose decay is entirely determined by tunneling probabilities. Thus it is desirable that approximate tunneling expressions can span the full range of tunneling from the bottom to the top of reaction barriers.

A clever and convenient approximate tunneling expression was first developed by Miller et al.⁵ more than twenty years ago. For the rest of this article, this expression will be referred to by the label MHHJW after the last names of its coauthors. It is a semiclassical, multidimensional analytic formula motivated by and based on second-order vibrational perturbation theory (VPT2). With VPT2, a reaction barrier is characterized not only

by its barrier height and imaginary frequency but also by its x matrix elements. In the intervening two decades, VPT2 has become more routine, especially for electronic structure methods with analytic gradients, and is now an option for several software packages such as Gaussian⁶ and CFOUR.⁷ Upon discovery of a reaction barrier and the associated harmonic force fields, a finite number of additional calculations that scale linearly with the degrees of freedom can obtain the cubic and quartic constants that supply the information needed to evaluate the x matrix.^{8–11} Each one of these additional calculations is independent of the others and thus the set of additional calculations can be done simultaneously in parallel on modern computers. Because the MHHJW formula exploits VPT2, the reaction kinetics software package Multiwell² incorporates this expression as its leading, but not exclusive, way of representing tunneling.

The MHHJW formula is a reliable approximation for thermal kinetics processes that heavily weigh tunneling through the top of the barrier. However, it has not been previously recognized that it is qualitatively incorrect for deep tunneling. This breakdown is due to the fact that VPT2 and therefore the formula do not fully exploit information that is always known in reaction barrier problems, namely the height of the barrier in both the forward and reverse direction. This article will show how such

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information can be used to improve the MHHJW formula for deep tunneling without affecting its representation of tunneling at the top of the barrier. The improved formula, although more complicated, remains analytic and retains the multidimensional aspect of the original formula. The basis of the improvement also sheds light on aspects of the VPT2 description of bound degrees of freedom perpendicular to the reaction path at the saddle point.

The rest of this article is presented as follows: Section 2 will briefly recapitulate the derivation of the MHHJW formula and describe the qualitative breakdown of the formula for deep tunneling and its origin. Section 3 will develop a way of correcting that breakdown while retaining an analytic expression. Section 4 discusses the implications of the correction for the bound degrees of freedom perpendicular to the tunneling coordinate. Section 5 is a summary.

2. DERIVATION AND LIMITATIONS OF THE MHHJW FORMULA

The MHHJW tunneling formula starts from the uniformly valid semiclassical tunneling probability¹² $P_{\text{SC}}(E)$ which is a function of the semiclassical tunneling action coordinate $\theta(E)$ where the energy E is measured from the reactant asymptote:

$$P_{\text{SC}}(E) = \frac{1}{1 + e^{2\theta(E)}} \quad (1)$$

The expression for $\theta(E)$ is determined from the VPT2 result:¹³

$$E(n_1, \dots, n_F) = V_1 + \sum_{k=1}^F \hbar\omega_k(n_k + 1/2) + \sum_{k=1}^F \sum_{k'=k}^F x_{kk'}(n_k + 1/2)(n_{k'} + 1/2) \quad (2)$$

where $F = 3N - 6$ for a nonlinear N -atom system, ω_k is the k th normal-mode frequency, $x_{kk'}$ is an element of the $\mathbf{x}$ matrix, and n_k is the quantum number for the k th mode. Equation 2 is the usual VPT2 expression for an equilibrium well, i.e., V_1 corresponds to the potential energy at the bottom of the well. When applied to a saddle point, i.e., V_1 is the barrier in the forward direction, one of the frequencies, which is designated the F th frequency, becomes imaginary and the off-diagonal elements in the corresponding column of the $\mathbf{x}$ matrix also become imaginary. The associated quantum number for the imaginary frequency becomes the action variable θ . This results in the following substitutions into eq 2:

$$\begin{aligned} \omega_F &= i|\omega_F| \\ n_F + 1/2 &= i\theta/\pi \end{aligned} \quad (3)$$

$$x_{kF} = ix_{kF}$$

for $k < F$. Upon substitution, eq 2 becomes quadratic in θ leading to an inverse of θ as a function of E in the form of:

$$\theta(n, E) = \frac{2\pi D}{\hbar\Omega_F} \left(1 - \sqrt{1 - \frac{\Delta E}{D}} \right) \quad (4)$$

where $n = (n_1, \dots, n_{F-1})$,

$$\begin{aligned} \Delta E &= V_1 + \sum_{k=1}^{F-1} \hbar\omega_k(n_k + 1/2) \\ &+ \sum_{k=1}^{F-1} \sum_{k'=k}^{F-1} x_{kk'}(n_k + 1/2)(n_{k'} + 1/2) - E \end{aligned} \quad (5)$$

$$\hbar\Omega_F = |\hbar\omega_F| - \sum_{k=1}^{F-1} |x_{kF}|(n_k + 1/2) \quad (6)$$

$$D = -\frac{(\hbar\Omega_F)^2}{4x_{FF}} \quad (7)$$

Equations 1 and 4 constitute the MHHJW formula for the tunneling probability.

In eq 4, the vector n dependence is found in ΔE , $\hbar\Omega_F$, and D , conferring on $\theta(E)$ its multidimensional character. The ΔE dependence on n is straightforward. If V_1 and E are measured from the *zero point corrected* base of the barrier on the reactant side, then ΔE is the difference between the total energy E and the top of the barrier plus the energy locked up, as defined by n , in degrees of freedom perpendicular to the reaction coordinate. $\hbar\Omega_F$ is the effective imaginary frequency corrected by anharmonic couplings of the reaction coordinate to perpendicular bound degrees of freedom that are a function of n . D acquires its n dependence from $\hbar\Omega_F$. From eq 7 D might appear to be negative but it is typically positive because x_{FF} is typically negative. x_{FF} is sensitive to the quartic and cubic derivatives of potential with respect to the reaction path normal mode as evaluated at the saddle point. These derivatives “flatten out” the parabolic barrier as the reaction path approaches reactant and product asymptotes. The parabolic limiting form of the barrier comes from the second derivative that alone does not lead to flat asymptotes.¹⁴

When written in the form of eq 4, the qualitative breakdown of the MHHJW formula for deep tunneling becomes evident. As E decreases, from eq 5 ΔE increases. If E becomes so small that ΔE exceeds D , then the action becomes complex and the tunneling formula of eq 1 is no longer physically interpretable. Since D has no direct connection to V_1 , at what energy this occurs has no direct connection to the energetics of the barrier. If D is less than V_1 , the tunneling for $0 \leq E \leq V_1 - D$ will be undefined. If D is greater than V_1 , there will be a finite tunneling probability for E below the base of the barrier.

The origins of D in eq 4 and the consequent breakdown of the MHHJW formula arise from its VPT2 origins. Equation 4 is not the semiclassical action for an energy approximated by eq 2 but for an energy *exactly* represented by eq 2. If one reduces eq 2 to a one-dimensional expression for only the F , or tunneling, degree of freedom and applies the substitutions of eq 3, the result is:

$$\Delta E = |\hbar\omega_F| \frac{\theta}{\pi} + x_{FF} \left(\frac{\theta}{\pi} \right)^2 \quad (8)$$

This is the Morse oscillator expression for ΔE for which VPT2 is exact.⁸ The well depth of this Morse oscillator is D and corresponds to the energy at which the energy level spacings of the oscillator “turn over” because the negative anharmonic second term in eq 8 becomes larger than the positive harmonic term. Equation 8 is measuring energy as ΔE or in other words down from the top of the barrier rather than more typically up from the bottom of a well. This implies that MHHJW formula is equivalent to semiclassical tunneling through an *inverted* Morse

oscillator barrier. If y is the reaction coordinate, then an inverted Morse oscillator barrier $V_{\text{IMO}}(y)$ has the form:

$$V_{\text{IMO}}(y) = D(2e^{\alpha y} - e^{2\alpha y}) \quad (9)$$

As pointed out by Miller et al., eq 4 is a multidimensional generalization of the barrier penetration integral expression for the semiclassical action. In one dimension that integral takes the form:

$$\theta(E) = \sqrt{\frac{2m}{\hbar^2}} \int \sqrt{V(y) - E} \, dy \quad (10)$$

where the limits on the integral are the turning points in the potential $V(y)$ that make the integrand go to zero and m is the reduced mass associated with motion along the reaction path at the saddle point. If $V(y) = V_{\text{IMO}}(y)$, then the integral is analytic with the resulting expression

$$\theta(E) = \pi \sqrt{\frac{2mD}{(\hbar\alpha)^2}} \left(1 - \sqrt{1 - \frac{\Delta E}{D}} \right) \quad (11)$$

As will be shown for other potentials in the next section, α can be adjusted to reproduce $|\hbar\omega_F|$ because this quantity is determined by the second derivative of $V(y)$ at the barrier which for $V_{\text{IMO}}(y)$ is proportional to α^2 . The elimination of the α dependence in eq 11 in favor of an $|\hbar\omega_F|$ dependence leads directly to eq 4 restricted to one dimension by $\hbar\Omega_F \rightarrow |\hbar\omega_F|$. In the typical implementation of a barrier penetration integral, D would be set by the actual barrier. However, through eq 7, D is set by x_{FF} to exploit the new information provided by VPT2. This breaks the direct connection of D to the energetics of the barrier. D is the barrier height *only* if ΔE is *exactly* described by eq 8, implying VPTn for $n > 2$ will yield zero coefficients for higher order corrections to the ΔE expression. There is no reason to expect this for the bond-breaking/bond-forming process at the saddle point of a realistic potential energy surface. Consequently the MHHJW formula is typically unreliable for deep tunneling. Another way to say this is that while V_1 , $|\hbar\omega_F|$, x_{FF} , and V_2 (the barrier in the endothermic reverse direction) are typically known, the inverted Morse oscillator has only two parameters (α , D). The MHHJW formula uses only $|\hbar\omega_F|$ and x_{FF} to set those to parameters in order to give the most accurate tunneling near the top of the barrier.

Note that the barrier penetration integral of eq 10 does not uniquely connect $\theta(E)$ to a $V(y)$. This is amply illustrated by the symmetric Eckart potential¹⁵ $V_{\text{SE}}(y)$

$$V_{\text{SE}}(y) = 4D \frac{e^{\alpha y}}{(1 + e^{\alpha y})^2} \quad (12)$$

Had this been substituted into eq 10, eq 4 would again result upon α elimination in favor of $|\hbar\omega_F|$. This result was commented on by Ahmed et al.¹⁶ V_{SE} and V_{IMO} are qualitatively different on the product side of the reaction path with V_{SE} having a product asymptote at the same energy of the reactant asymptote but V_{IMO} having a product asymptote infinitely below the energy of the reactant asymptote. Nonetheless they have the same semiclassical tunneling probability if their two parameters are the same.

3. IMPROVED SEMICLASSICAL TUNNEL THEORY

The challenge of improving the MHHJW formula is to preserve its correct sensitivity to $|\hbar\omega_F|$ and x_{FF} while analytically but fully incorporating information always known in tunneling problems,

namely V_1 and V_2 . (In the rest of this paper, the “forward” direction of the reaction is the exoergic direction with a barrier V_1 while V_2 is the barrier for the reverse endoergic direction.) The approach developed here is to develop a suitably flexible potential $V(y)$ that incorporates all the known information but still has an analytic barrier penetration integral. Part 3.1 will develop the potential, Part 3.2 will describe the analytic action variable θ , and Part 3.3 will examine the corresponding $P_{\text{SC}}(E)$. Each part of this section will briefly motivate and discuss the appropriate equations and refer to the Supporting Information for the full and self-contained derivations. For convenience, this section will use the language of a one-dimensional barrier. The multidimensional aspect of the MHHJW formula can be recovered by replacing everywhere $|\hbar\omega_F|$ by $\hbar\Omega_F$ and by interpreting the energy E as the total energy minus the energy in the bound degrees of freedom described as in eq 5. This will be further discussed in Section 4.

3.1. Potential. To incorporate all the known information, a potential with at least four adjustable parameters is required. There are relatively few chemically relevant potentials with analytic barrier penetration integrals.¹⁷ Most have two parameters, as in eqs 9 and 12, but the asymmetric Eckart potential does have three parameters:

$$V(y) = V_1 \left((1 - \rho^2) \frac{e^{\alpha y}}{(1 + e^{\alpha y})} + (1 + \rho)^2 \frac{e^{\alpha y}}{(1 + e^{\alpha y})^2} \right) \quad (13)$$

where $\rho = (V_2/V_1)^{1/2}$. Inspection will show that eq 13 reduces to V_{SE} of eq 12 if $\rho = 1$ and $V_1 = D$. With only three adjustable parameters, one can hope to fit $|\hbar\omega_F|$, x_{FF} , and either V_1 or V_2 . Since V_2 is not as essential as fitting V_1 , optimizing the three parameters by sacrificing an accurate value of V_2 in favor of the correct value of x_{FF} seems an acceptable, if not ideal, strategy. Unfortunately, this strategy frequently is *not* acceptable because the resulting V_2 is often less than V_1 , meaning that tunneling in the V_1 direction is *endoergic*, precluding the ability to compute $P_{\text{SC}}(E)$ for $0 \leq E \leq V_1 - V_2$. Of course, in the accidental case that D of eq 5 also just happens to be the true barrier V_1 , then eq 13 will reproduce all four available pieces of information and the resulting action and consequent tunneling probabilities will not suffer from the problems discussed in Section 2. This unique case will not be covered in this article except to point out in Part 3.2 of this section that the resulting action will not be identical to eq 4 if $\rho \neq 1$, i.e., an asymmetric Eckart potential instead of the symmetric one of eq 12.

The necessary flexibility in the potential can be provided by a composite of different Eckart potentials matched in value and slope at selected values of y . Since any individual Eckart potential has an analytic barrier penetration integral, a composite will also. Continuity of the composite potential in value and slope at the match points will also guarantee continuity of θ in value and slope at the energy of the match points. However, in the absence of further information, there is no unique way to select the match points. This means that the composite Eckart approach will ultimately produce probabilities with some degree of arbitrariness that cannot be removed without additional explicit information about the true reaction path. In what follows, a three-segment Eckart composite potential matched together at two selected values of y is described. This composite correctly reproduces $|\hbar\omega_F|$, x_{FF} , V_1 , and V_2 . The principles behind the selection of the two match points are reasonable and the resulting θ and corresponding $P_{\text{SC}}(E)$ are well behaved. After this

description, alternative methods of selecting match points are briefly discussed.

From a minimalist point of view, it would be desirable for each of the three potentials in our composite to share a common and correct value for at least one of the three parameters V_1 , ρ , and α in an Eckart potential. This can be achieved by fixing ρ to its correct value of $(V_2/V_1)^{1/2}$. Then the three segments are separated into a reactant segment, a barrier segment, and a product segment. For the barrier segment V_b , the V_1 and α parameters are set to the unique values D_b and α_b that correctly reproduce $|\hbar\omega_F|$ and x_{FF} . (How to determine D_b and α_b will be discussed after the description of all three segments.) The whole potential is then shifted in energy by $V_1 - D_b$ to give:

$$V_b(y) = V_1 - D_b + D_b \left((1 - \rho^2) \frac{e^{\alpha_b y}}{(1 + e^{\alpha_b y})} + (1 + \rho)^2 \frac{e^{\alpha_b y}}{(1 + e^{\alpha_b y})^2} \right) \quad (14)$$

This energy-shifted potential has no effect on the analyticity of the barrier penetration integral. Because of the shift, this potential does describe the region of the barrier correctly but has incorrect reactant and product asymptotes. These incorrect asymptotes are corrected by the reactant V_r and product V_p segments, both of which use the correct values of V_1 and ρ but adjust α to α_r and α_p , respectively, to match to V_b at the appropriate match points. The end result is a composite with the correct energetics and a modified barrier region that correctly reproduce $|\hbar\omega_F|$ and x_{FF} .

To implement this three-segment composite potential, D_b and α_b must first be determined, followed by α_r and α_p , and then the appropriate match points. These will each be discussed in turn.

The value of α_b is primarily set by $|\hbar\omega_F|$. At the top of the V_b barrier, the second derivative of V_b with respect to y determines the harmonic force constant in the expansion of V_b about the barrier in the reduced normal mode q along the reaction path, as in

$$\frac{1}{2} \left| \frac{\partial V_b}{\partial y^2} \right| (y - y_{mx})^2 = \frac{1}{2} \left(\frac{\hbar}{m|\hbar\omega_F|} \left| \frac{\partial V_b}{\partial y^2} \right| \right) q^2 = \frac{1}{2} |\hbar\omega_F| q^2 \quad (15)$$

where y_{mx} is the barrier location and m is the reduced mass associated with the normal mode for the tunneling. As discussed in detail in the Supporting Information (see eq S1.5), the second derivative of V_b with respect to y is proportional to α_b^2 . Via eq 15 this leads to the defining equation for α_b :

$$\alpha_b = |\hbar\omega_F| \frac{(1 + \rho)}{\rho} \sqrt{\frac{m}{2\hbar^2 D_b}} \quad (16)$$

While the value of m is necessary to map V_b as a function of y , the action and consequent probability determined from the potential will ultimately not depend on m . Consequently in discussing and displaying the composite potential, m will be assigned a nominal value of 1.0 amu.

The value of α_b in eq 16 requires the value of D_b . That value is primarily set to reproduce x_{FF} . For an effective one-dimensional potential, that relationship is from Miller et al.⁵

$$x_{FF} = -\frac{1}{16} \left(\frac{\partial^4 V}{\partial q^4} + \frac{5}{3} \frac{\left(\frac{\partial^3 V}{\partial q^3} \right)^2}{|\hbar\omega_F|} \right) \quad (17)$$

where q is again the reduced normal mode of the reaction coordinate (Miller et al. expressed the derivatives in the unreduced mass weighted normal mode). As detailed in the Supporting Information (see eq S1.9), the evaluation of these derivatives for V_b followed by elimination of α_b in favor of an $|\hbar\omega_F|$ dependence leads to the expression

$$D_b = D \left(1 - \frac{1}{\rho} + \frac{1}{\rho^2} \right) \quad (18)$$

where, as discussed before, D of eq 7 characterizes the breakdown in the MHHJW formula and is composed from both $|\hbar\omega_F|$ and x_{FF} . Equations 14, 16, and 18 define V_b .

The physical picture that motivates the determination of α_r and α_p , and also the appropriate match points, is best visualized by expressing the derivative of an Eckart potential V not as a function of y but as a function of $\Delta = V_1 - V(y)$, i.e., the difference between the maximum and current value of the potential. (Δ is for potential energy the analogue of ΔE .) The same can also be done for V_r and V_p . As derived in the Supporting Information for V_r (see eq S2.4) and generalized here, the form of the Eckart potential derivative expressed as a function of Δ is:

$$\frac{\partial V_s}{\partial y} = \mp \frac{2\alpha_s V_{1s}}{1 + \rho} \sqrt{\frac{\Delta}{V_{1s}}} \left[\rho \mp (1 - \rho) \sqrt{\frac{\Delta}{V_{1s}}} - \frac{\Delta}{V_{1s}} \right] \quad (19)$$

where when $s = b$, $\alpha_s = \alpha_b$, and $V_{1s} = D_b$ while when $s = r(p)$, $\alpha_s = \alpha_{r(p)}$ and $V_{1s} = V_1$. The two choices of signs in eq 19 identify the reactant (+ sign) or product (− sign) side branch of the derivative. In this expression, Δ can only physically range from 0 to V_{1s} and thus the range of Δ is different between V_b and $V_{r(p)}$. The two solid lines in Figure 1 display $\partial V_s/\partial y$ for $s = b$ and r where $\rho = 1.1$ and $\alpha_r = \alpha_b = 1.0$. As would be expected for the reactant side of a potential with a barrier, the slope is zero at the asymptote ($\Delta = V_{1s}$) and at the barrier ($\Delta = 0$) with a maximum

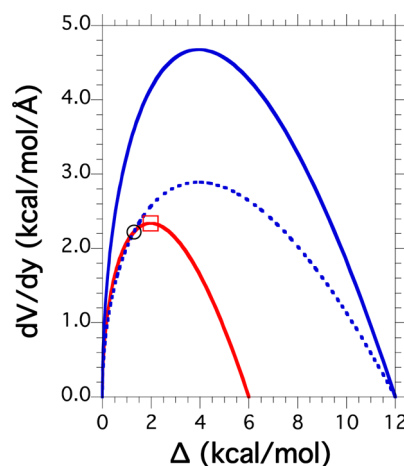


Figure 1. For $r = 1.1$, $\partial V_s/\partial y$ versus Δ for (s, V_{1s}, α_s): ($r, 12$ kcal/mol, 1.0 Å^{-1}) solid blue line, ($r, 12$ kcal/mol, 0.618 Å^{-1}) dotted blue line, and ($b, 6$ kcal/mol, 1.0 Å^{-1}) solid red line. See text for details. The black circle identifies the intersection point while the red square indicates the maximum in the slope for V_b (see text).

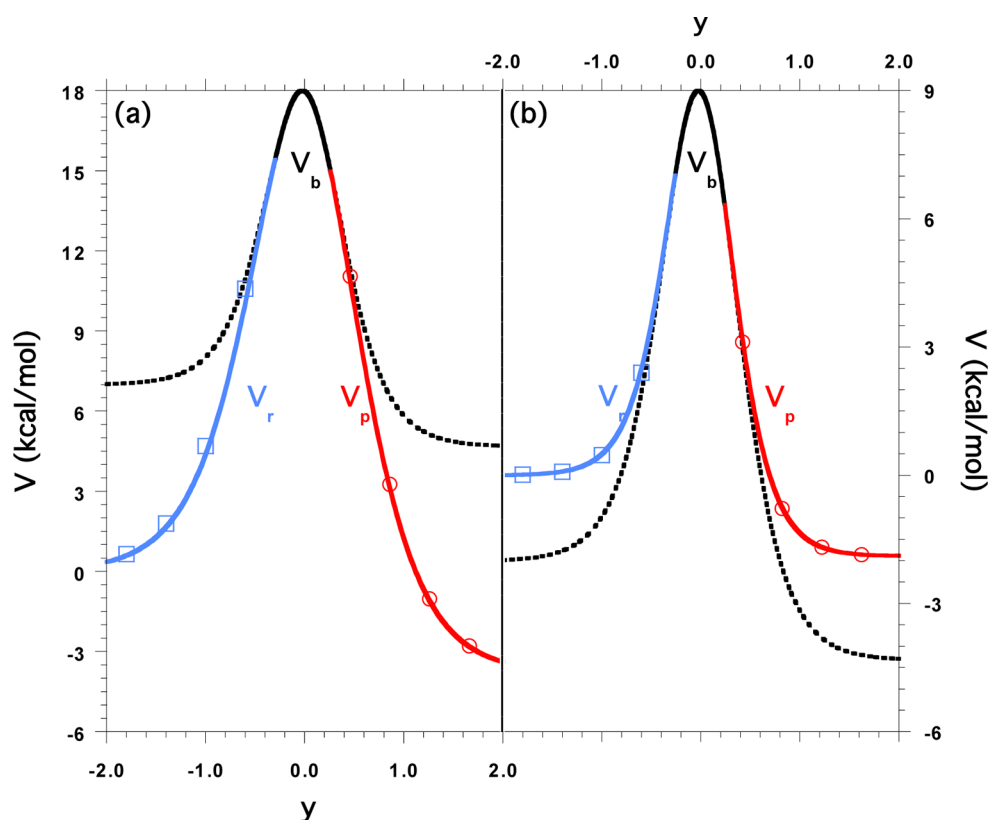


Figure 2. (a) For the case $V_1 = 18$ kcal/mol, $r = 1.1$, $|\hbar\omega_P| = 1000$ cm $^{-1}$, and $D = 12$ kcal/mol, the three-segment composite potential as a function of y . V_b is the black solid line, V_r is the blue solid line, and V_p is the red solid line. The dotted black line is V_b beyond the match points (see text). The square and circle symbols indicate the location of the *unshifted* V_r and V_p , respectively (see text). (b) As in panel a for $V_1 = 9$ kcal/mol.

positive value at the inflection point in between (an open square locates the inflection point for V_b in Figure 1). A plot of the product side of the potential would look similar only with $\partial V_p/\partial y \leq 0$.

Figure 1 and eq 19 make clear what the match of V_b and V_r must accomplish. From eq 19, the size of $\partial V_r/\partial y$ at any value of Δ is directly proportional to α_r . From Figure 1, for $\alpha_r = \alpha_b$, the $\partial V_r/\partial y$ curve for whichever is the larger of V_b and V_r will always encompass, but not intersect, the $\partial V_b/\partial y$ curve for whichever is the smaller of V_b and V_r . If α_r is altered so that the two curves intersect, then at the intersection point, V_b and V_r will have a common value of the slope at a common potential energy below the barrier where the barrier by construction has a common value for both curves. The resulting composite potential matched at the intersection point will be continuous in value and slope on the reactant side and, by analogous operations on the product side, be continuous in value and slope everywhere. The only technical complication in this approach is that all three components of the composite potential have a y location of the barrier given by $-\ln(\rho)/\alpha_s$. Since this strategy will make $\alpha_{r(p)} \neq \alpha_b$, $V_{r(p)}$ must be shifted in y space to achieve a match with V_b . If the shift is labeled $d_{r(p)}$, from eq 13, $V_{r(p)}$ assumes the form

$$V_{r(p)}(y) = V_1 \left[(1 - \rho^2) \frac{e^{\alpha_{r(p)}(y+d_{r(p)})}}{(1 + e^{\alpha_{r(p)}(y+d_{r(p)})})} + (1 + \rho)^2 \frac{e^{\alpha_{r(p)}(y+d_{r(p)})}}{(1 + e^{\alpha_{r(p)}(y+d_{r(p)})})^2} \right] \quad (20)$$

To complete the description of $V_{r(p)}$, $\alpha_{r(p)}$ and $d_{r(p)}$ must be defined. $\alpha_{r(p)}$ is determined from a least-squares fit to minimize

$(\partial V_r/\partial y - \partial V_b/\partial y)^2$ over the range of Δ from 0 to its value at the inflection point. As derived in the Supporting Information (see eq S2.5) for the reactant side and generalized here, the value of Δ at the inflection point, labeled Δ_{IP} , is independent of α_s and can be expressed as:

$$\Delta_{IP} = \frac{V_{1s}}{9} [\sqrt{\rho^2 + \rho + 1} \pm (\rho - 1)]^2 \quad (21)$$

The upper sign is for the product side and the lower sign is for the reactant side. Since matching involves two curves of different slopes for each side of the potential, there are two inflection points and hence two values of Δ_{IP} for each potential. The smaller one is always the most appropriate, i.e., V_{1s} in eq 21 is replaced by $\min(V_1, D_b)$. For Δ values as high as the smaller Δ_{IP} , the slope of V_b is clearly inconsistent with the true energetics that the composite potential will reflect. In Figure 1 where $D_b < V_r$, the slope is not large enough to lead to V_b decaying to the much lower asymptote of V_r . Were $D_b > V_r$, the slope would be too large to lead to V_b decaying to the higher asymptote of V_r . The least-squares process incorporates this physical picture.

In Figure 1, the dotted line is $\partial V_r/\partial y$ for the resulting value of α_r that comes from this least-squares process (the open square in Figure 1 indicates the Δ_{IP} used). The least-squares process always leads to an intersection of the two curves. The composite potential will have a slope identical to $\partial V_b/\partial y$ for $\Delta \leq \Delta_{IP}$ and a slope identical to $\partial V_{r(p)}/\partial y$ for $\Delta \geq \Delta_{IP}$. The scale of the unavoidable discontinuity in the second derivative of the composite potential is clear from the angle of intersection of the two curves in Figure 1. As shown in the Supporting Information (see eqs S2.6, S2.7, and S2.9), the least-squares process is completely analytic resulting in a formula for $\alpha_{r(p)}$.

That formula shows $\alpha_{r(p)}$ to be proportional to $\alpha_b(D_b/V_{r(p)})^{1/2}$ with a proportionality constant that is the ratio of two polynomials in Δ_{IP} .

The remaining value of $d_{r(p)}$ is determined by the shift in $V_{r(p)}$ needed to obtain a match in the value of the potential at the intersection of the two $\partial V/\partial y$ curves that occurs in the least-squares determination of $\alpha_{r(p)}$. In Figure 1, the intersection is indicated by the open circle. As derived in the Supporting Information (see eqs S3.11–S3.14), $\Delta_{m,r(p)}$, the value of Δ at the intersection, can be calculated from an analytic expression and from that expression the values of $y_{br(p)}$ and $y_{r(p)b}$ where $V_1 - V_b(y_{br(p)}) = V_1 - V_{r(p)}(y_{r(p)b}) = \Delta_{m,r(p)}$ can be analytically determined. Then, $d_{r(p)} = y_{br(p)} - y_{r(p)b}$. That there is a displacement at all is due to the fact that $\alpha_{r(p)}$ and α_b are not identical.

Equation 20 and eqs S2.7 to S2.22 in the Supporting Information define the segments $V_{r(p)}$. Together with V_b defined earlier, they complete the description of a three-segment composite Eckart potential that correctly represents V_1 , V_2 , $|\hbar\omega_F|$, and x_{FF} and will have an analytic barrier penetration integral (as discussed in the next part of this section). Figure 2 displays this composite potential for identical values of ρ , $|\hbar\omega_F|$, and x_{FF} (as defined by D) but two different values of V_1 (18 kcal/mol in Figure 2a and 9 kcal/mol in Figure 2b). The composite potential is smooth and well behaved everywhere in both figures. As indicated in Figure 2a, the reactant asymptote for V_b is higher than the correct asymptote. The degree that $V_{r(p)}$ needs to be shifted by $d_{r(p)}$ is quite small and can only be distinguished on the scale of the plot by the slightly displaced square symbols for V_r . In Figure 2b, the reactant asymptote for V_b is lower than the correct asymptote and the degree of shifting is even smaller.

Figure 3 shows a family of six composite potentials, all with the same values of ρ , $|\hbar\omega_F|$, and x_{FF} (as defined by D) used for Figure

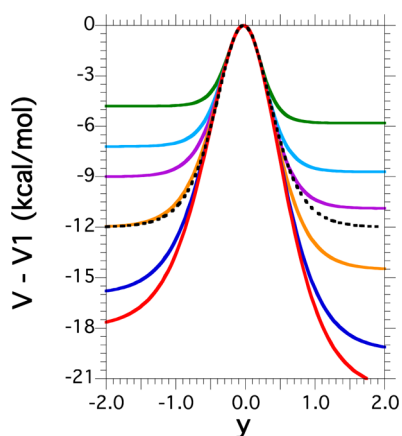


Figure 3. For the case $r = 1.1$, $|\hbar\omega_F| = 1000 \text{ cm}^{-1}$, and $D = 12 \text{ kcal/mol}$, a family of composite potentials where $V(y) - V_1$ is plotted as a function of y where (color, V_1 in kcal/mol) = (green, 4.8), (sky blue, 7.2), (magenta, 9), (orange, 12), (blue, 16), (red, 18). The dotted black line is a shifted symmetric Eckart potential with a barrier height of D (see text for details).

2 but with values of V_1 that range from 4.8 to 18 kcal/mol. The ordinate of the plot is $V - V_1$, making all six potentials overlap at the location of the barrier. Two of the composite potentials are those in Figure 2. All six composite potentials overlap over a range that encompasses the barrier as by construction they must. The dotted line is the specific symmetric Eckart potential of eq 12 whose barrier penetration integral produces exactly the same

action as that for the MHHJW formula. (For clarity, this potential has been slightly shifted in the figure so that its maximum occurs at the common maximum of the six composite potentials.) As Figure 3 shows, this symmetric Eckart potential overlaps the six composite potentials in the barrier region as indeed it must because all the potentials in Figure 3 reproduce the same $|\hbar\omega_F|$ and x_{FF} . This means that the MHHJW formula approximates the tunneling of all the potentials in Figure 4 as being that of the

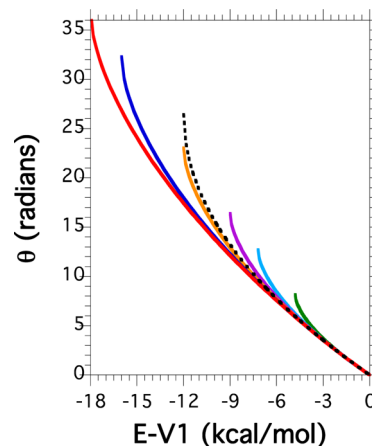


Figure 4. The color-coded action θ as a function of $E - V_1$ for the appropriately colored potential in Figure 3. The dotted line is θ from the MHHJW formula for the common value of $|\hbar\omega_F|$ and D shared by all the potentials.

tunneling for the symmetric Eckart potential in the figure. Thus Figure 3 is a visual representation of the fact that the MHHJW formula is qualitatively incorrect for deep tunneling. As will be shown in the next part of this section, each composite potential has an analytic barrier penetration integral and thus forms the basis for an improved semiclassical formula that reflects the differences of each potential in Figure 3.

As discussed at the start of this part of the section, there is no unique composite Eckart potential. However, some composites have ancillary undesirable properties for this problem. As an illustration, consider an apparently simpler two-segment composite Eckart potential matched in value, derivative, and second derivative at the location of their respective barriers. On the reactant (product) side of the barrier, $V_{1(2)}$ is fixed at the correct value while ρ and α are adjusted to reproduce $|\hbar\omega_F|$, and x_{FF} . As in the three-segment composite, the two potentials will be displaced from one another in y space and one will have to be shifted to match the other at the barrier. This two-segment model has one less segment, a simpler matching criterion, and is continuous everywhere to a higher derivative of the potential than the three-segment model just discussed. However, it has two serious disadvantages. First, for V_1 or $V_2 < 0.75D$, there is no real value of ρ that can match x_{FF} . This can be seen from eq 18 where if D_b is replaced by $V_{1(2)}$ in the formula, it can be seen that V_1/D is bounded from below. This can be cured only by breaking the reactant and perhaps product segment into two segments where an inner segment uses an adjustable $V_{1(2)}$ to ensure a correct x_{FF} and an outer segment uses the correct $V_{1(2)}$ to restore the true energetics. Thus the two segment composite strategy can in fact require up to four segments and corresponding matching condition complexity. The second disadvantage is that high order discontinuities in the potential at the barrier introduce for all relevant energies terms in the action that are not present in the

MHHJW formula for the action. As is discussed in the next part of this section and developed in the Supporting Information, these additional terms cause the resulting action to deviate from the MHHJW formula by the second order in ΔE .

In contrast the three-segment composite potential model developed in this part has to second order in ΔE the same action as the original MHHJW formula until $E - V_1$ drops below $\Delta_{m,r(p)}$ and the correction for the energetics begins to take hold. The three-segment composite model always has three segments, never four, no matter what the relationship between V_1 and D . However, the precise nature of the matching of the segments in effect controls the energy region over which the MHHJW tunneling formula applies. While reasonable, the matching is not unique and cannot be made so without additional information on the actual potential.

3.2. Action. Given the three-segment composite potential described above, the action $\theta(E)$ expressed in terms of the barrier penetration integral assumes the form:

$$\theta(E) = \sqrt{\frac{2m}{\hbar^2}} \left\{ \int_{y_L}^{y_{br}} \sqrt{V_r(y) - E} dy + \int_{y_{br}}^{y_{bp}} \sqrt{V_b(y) - E} dy + \int_{y_{bp}}^{y_H} \sqrt{V_p(y) - E} dy \right\} \quad (22)$$

where $y_{L(H)}$ is the reactant (product) side turning point at energy E for $V_{r(p)}(y)$. When E is near the barrier, $y_L > y_{br}$ and $y_H < y_{bp}$ reducing $\theta(E)$ to only the middle integral with limits y_L and y_H . As indicated in eq 20 for $V_{r(p)}$, the first and third integrals are of the same form but evaluated with different parameter values. As indicated in eq 12, V_b has the same form as $V_{r(p)}$ only with an energy shift of $V_1 - D_b$. That energy shift can be subtracted from E in the middle integral. Then, as shown in the Supporting Information (see eqs S3.3–S3.7), eq 22 becomes a linear combination of three integrals of a common form I . Expressed as an indefinite integral, I is

$$I = \int \frac{\sqrt{-\eta_2 z^2 + 2z\eta_\rho - \eta}}{z(1+z)} dz \quad (23)$$

where $\eta = E/V_1$ for $V_{r(p)}$ and $[E - (V_1 - D_b)]/V_1$ for V_b and where $\eta_\rho = 1 + \rho - \eta$ and $\eta_2 = \eta + \rho^2 - 1$. This integral is analytic:

$$I = (1 + \rho) \arcsin\left(\frac{\rho z - 1}{(1 + \rho)\sqrt{1 - \eta}}\right) - \sqrt{|\eta|} \left\{ \begin{array}{ll} \arcsin\left(\frac{\eta_\rho z - \eta}{z(1 + \rho)\sqrt{1 - \eta}}\right) & \text{if } \eta > 0 \\ \ln\left(\frac{-\eta + \eta_\rho z + \sqrt{\eta(\eta - 2\eta_\rho z + \eta_2 z^2)}}{z}\right) & \text{if } \eta < 0 \end{array} \right\} + \sqrt{|\eta|} \left\{ \begin{array}{ll} \arcsin\left(\frac{\eta_\rho - \eta_2 z}{(1 + \rho)\sqrt{1 - \eta}}\right) & \text{if } \eta_2 > 0 \\ \ln(\eta - \eta_2 z + \sqrt{\eta_2(\eta - 2\eta_\rho z + \eta_2 z^2)}) & \text{if } \eta_2 < 0 \end{array} \right\} \quad (24)$$

When evaluated at the appropriate limits specific to the three integrals in eq 22 and multiplied by the correct coefficients in the linear combination (see eq S3.6 in the Supporting Information), an analytic expression for $\theta(E)$ results.

While analytic, the resulting formula for $\theta(E)$ is considerably more complicated than the MHHJW formula of eq 4. However,

it is not as complicated as eq 24 would initially suggest. To begin with, η and η_2 in eq 24 can be expressed as fractions that differ only in the numerator. For η , the numerator is the energy above the reactant asymptote while for η_2 the numerator is the energy above the product asymptote. It is not possible for E to be below either asymptote for the first and third integral in eq 22 because $V_{r(p)}$ has the correct energetics. It is not possible for E to be below the asymptote of V_b if D_b is larger than V_1 or if E is above $V_b(y_{br(p)})$. In such cases, only the arcsin terms in eq 24 apply. In this case, as is shown in the Supporting Information (see eqs S3.10 and S3.11), when evaluated at a turning point, i.e., $y_{L(H)}$, the arguments of all three arcsin expressions are either +1 or -1 and eq 24 reduces to a simple expression. At high enough energy, $y_L > y_{br}$ and $y_H < y_{bp}$, only the middle integral of eq 22 applies, and that integral is evaluated at the turning points. As shown in the Supporting Information (see eq S3.12), the resulting simple expression for $\theta(E)$ is:

$$\theta(E) = \frac{2\pi D_b}{|\hbar\omega_F|} \frac{\rho}{1 + \rho} \left(1 + \rho - \sqrt{\frac{E - (V_1 - D_b)}{D_b}} - \sqrt{\frac{E - (V_1 - \rho^2 D_b)}{D_b}} \right) \quad (25)$$

The two numerators that are functions of E in eq 25 are the energy above the V_b reactant asymptote and the product asymptote (see eq 14). If the Eckart potential is symmetric, i.e., $\rho = 1$, and there is no energy shift in V_b , i.e., $D_b = V_1$, then eq 25 reduces in form exactly to eq 4, the MHHJW expression for the action. While exact in form, it becomes exact in value only when $D_b = D$, confirming the discussion concerning eq 12 that both a symmetric Eckart and an inverted Morse oscillator have the action of eq 4 when the barrier is set to D . If expanded about $E = V_1$ eqs 4 and 25 are identical to second order in $E - V_1$ as shown in the Supporting Information (see eq S3.14). This guarantees that the improved action of eqs 22–24 merge smoothly into the MHHJW action as the energy approaches the vicinity of the barrier.

In Figure 4, the improved semiclassical $\theta(E)$ is computed by using eqs 22–24 for all the potentials in Figure 3 including the symmetric Eckart potential for which $\theta(E)$ is identical to original action formula of eq 4. As the figure indicates and as expected, all of the potentials in Figure 3 produce actions that coalesce as E approaches the top of the barrier. Figure 4 also shows that the improved semiclassical actions do properly terminate when E reaches zero at the bottom of the barrier on the reactant side. As discussed in Section 1, the MHHJW action does not do that because it has no information on where the bottom of the barrier is. The MHHJW action formula interprets D as the energetic information producing a single $\theta(E)$ that represents all the actions in Figure 4 and that can be derived from the single symmetric Eckart potential in Figure 3.

Equation 25 by itself is superior to the MHHJW action formula because it reflects the asymmetry of the potential by virtue of being an explicit function of ρ . One might expect its energy region of validity to be somewhat larger than the MHHJW action. In the context of eqs 22–24, its region of validity extends to that energy at which $y = y_{br(p)}$. While superior to the MHHJW action, eq 25 was not derived from a VPT2 expansion of ΔE in terms of action as in eq 8 for the MHHJW action. However, as shown in the Supporting Information (see eqs S5.1–S5.4), eq 25

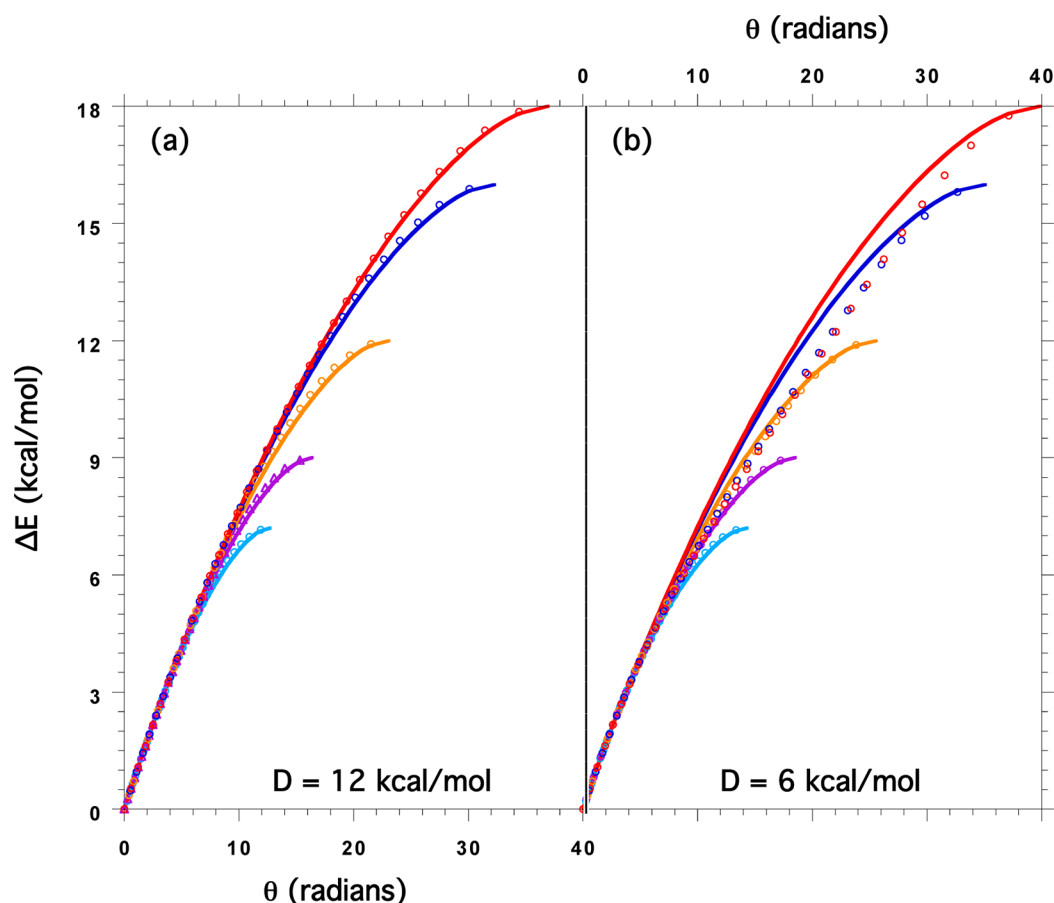


Figure 5. (a) ΔE versus θ as solid lines color-coded as in Figure 3 to indicate the potential from which each line was calculated. The corresponding color-coded open circles trace the quartic polynomial expansion of ΔE in θ (see text for details). (b) As in panel a only the common value of D used in panel a has been halved.

can be inverted into an analytic expression for ΔE in terms of the action:

$$\Delta E = |\hbar\omega_F| \frac{\theta}{\pi} + x_{FF} \left(\frac{\theta}{\pi} \right)^2 - \frac{2\rho(1-\rho)^2}{(1-\rho+\rho^2)^2} \left(\frac{x_{FF}}{|\hbar\omega_F|} \right) x_{FF} \left(\frac{\theta}{\pi} \right)^3 + \frac{(1-\rho)^2}{\rho} \sum_{n=4}^{\infty} (n+1)(-2)^{n-4} \left(\frac{\rho}{1-\rho+\rho^2} \right)^{n-1} \left(\frac{x_{FF}}{|\hbar\omega_F|} \right)^{n-2} x_{FF} \left(\frac{\theta}{\pi} \right)^n \quad (26)$$

When $\rho = 1$ and the Eckart potential is symmetric, eq 26 reduces to eq 8 as it should. In terms of vibrational perturbation theory, each even value of “ n ” in VPTn will produce a term of order $\theta^{(n+2)/2}$ in the expression for ΔE . Thus to recover the first term in eq 26 that contains information about ρ , i.e., the term cubic in θ , will require VPT4. However, any higher order n th term in the equation goes as $(x_{FF}/|\hbar\omega_F|)^{n-2}$. For realistic potentials the ratio will be quite small and raising the ratio to ever higher powers will cause the infinite summation to die off quickly. Consequently, high order vibrational perturbation theory much beyond VPT4 will provide minimal improvement in the description of $\Delta E(\theta)$.

Equation 26 concerns only the middle segment of the three-segment composite potential developed in this section. The two other segments are necessary to correctly reflect the barrier V_1 that the middle segment does not. Equation 26 shows that it takes at least VPT4 to recover information on ρ . One might

imagine that to recover information on both ρ and V_1 would require at least VPT6. If an analytic expression for $\theta(\Delta E)$ is desired, the inversion of $\Delta E(\theta)$ to $\theta(\Delta E)$ is still possible up to fourth order. As eq 26 suggests, the reliable recovery of both ρ and V_1 will undoubtedly require quite high order vibrational perturbation theory. Without doing any VPTn calculations, it is still known that $\Delta E(\theta)$ will have the value V_1 at its first maximum for positive values of θ . Exploiting these observations, one could expand ΔE to fourth order in θ such the first and second order coefficients are exact from VPTn and therefore reproduce ω_F and x_{FF} while the third and fourth order coefficients are not from VPTn but assume effective values that cause ΔE to peak at V_1 . The two higher order coefficients are set by two conditions only involving V_1 , namely that at a common value of $\theta = \theta_{mx}$ the slope of ΔE is zero while its value is V_1 . There is no way, short of VPTn calculations, of introducing into this approach information about ρ . In a sense, ρ is encoded in the value of θ at which the maximum in ΔE occurs. Furthermore, without a way to set the value of θ at the peak, a broad range of expansions are possible which ultimately lead to a large variation in possible tunneling probabilities.

This ambiguity can be eliminated by using eq 22 to obtain θ_{mx} i.e., evaluating eq 22 at $E = 0$. With that value of θ_{mx} one can test the ability of only a fourth-order expansion of $\Delta E(\theta)$ to describe the deep tunneling regime. The resulting expression for $\Delta E(\theta)$ is:

$$\begin{aligned} \Delta E = & |\hbar\omega_F| \left(\frac{\theta}{\pi} \right) + x_{FF} \left(\frac{\theta}{\pi} \right)^2 - \left[3|\hbar\omega_F| \left(\frac{\theta_{mx}}{\pi} \right) \right. \\ & + 2x_{FF} \left(\frac{\theta_{mx}}{\pi} \right)^2 - 4V_1 \left. \left(\frac{\theta}{\theta_{mx}} \right)^3 \right] \\ & + \left[2|\hbar\omega_F| \left(\frac{\theta_{mx}}{\pi} \right) + x_{FF} \left(\frac{\theta_{mx}}{\pi} \right)^2 - 3V_1 \right] \left(\frac{\theta}{\theta_{mx}} \right)^4 \end{aligned} \quad (27)$$

which satisfies $\Delta E(\theta_{mx}) = V_1$ and $\partial\Delta E(\theta_{mx})/\partial\theta = 0$. In Figure 5, eq 27 is applied to a variety of potentials with general but not complete success. Figure 5a indicates quite good agreement of eq 27 with the potentials in Figure 3. All of these potentials had a common value of D , $|\hbar\omega_F|$, and ρ . Figure 5b shows that reducing the value of D thereby increasing the value of x_{FF} makes the quartic expansion increasingly insufficient as V_1 increases. Other variations in $|\hbar\omega_F|$ and ρ from that of Figure 5a do not appear to significantly degrade the accuracy of the quartic expansion of eq 27. If one could determine θ_{mx} without reference to a constructed potential, eq 27 could be inverted to produce $\theta(\Delta E)$ without ever “bothering” to develop a potential. However, eq 22 evaluated at $E = 0$ does not produce a simple expression because all three integrals are involved and matching requirements complicate the expression (see eq S3.5 in the Supporting Information).

As an alternative to eq 27, the same strategy of a triple segmentation of the potential described in Part 3.1 could lead to a double segmentation of eq 8 without resorting to a potential at all. The segment at lower values of θ would be eq 8 unchanged but the higher segment would involve a correlated change of both the linear and quadratic coefficients of powers of a shifted θ . The correlation is set by assigning V_1 to the peak value of ΔE of the segment, leaving one parameter to adjust. The resulting free parameter is set by minimizing in a least-squares sense the difference of $\partial\Delta E/\partial\theta$ for the two segments where $\partial\Delta E/\partial\theta$ is expressed as a function of ΔE and the minimization is from 0 to $\min(D, V_1)$. The analytic minimization always results in an adjusted parameter value that leads to an intersection of the two segments. The analytic intersection or match value for ΔE then defines the shift in θ for the second segment. The result will be a ΔE as a function of θ that is continuous in slope and value from 0 to V_1 . That function can be inverted to give $\theta(\Delta E)$ with a form that will be about as simple as eq 10 only with parameters, i.e., D and α in eq 10, that change values at the match value of ΔE . However, the resulting expression will contain no information about ρ , the asymmetry of the potential. In Figure 4, the dotted line is the value of $\theta(E)$ for a symmetric, i.e., $\rho = 1$, Eckart potential with a barrier of 12 kcal/mol. The solid line for $V_1 = 12$ kcal/mol is a three-segment asymmetric, i.e., $\rho = 1.1$, Eckart potential with the same barrier. Both potentials also have identical values of $|\hbar\omega_F|$ and x_{FF} . The 15% difference in the θ_{mx} value between the two potentials is a reflection the 10% change in the value of ρ . In effect, the deeper product asymptote shaves off some of the barrier on the product side, leading to a lower action. This behavior would not be retained by a direct two-segment construction of $\theta(\Delta E)$ without recourse to prior construction of the potentials.

3.3. Probability. Given eq 1 and $\theta(E)$ from eqs 22–24, the tunneling probability $P_{SC}(E)$ can be readily obtained. Figure 6 displays $P_{SC}(E)$ for the potentials in Figure 3 which produced the $\theta(E)$ in Figure 4. The dotted black line is the MHHJW probability that would be applicable to all the cases in the figure because they all share common values for $|\hbar\omega_F|$ and x_{FF} . As the

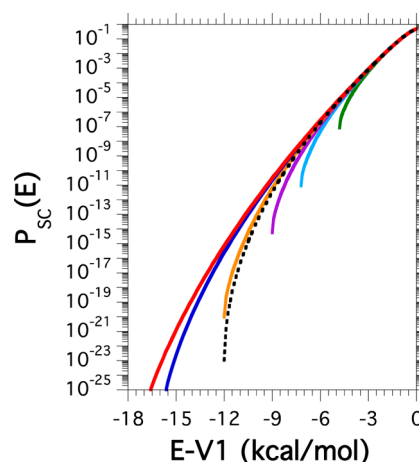


Figure 6. $P_{SC}(E)$ versus $E - V_1$ color-coded according to the potentials in Figure 3 from which the probabilities are derived.

figure shows, for about the first 3 kcal/mol below the barrier, all of the probabilities are identical on the scale of the figure. At energies further below the barrier, the improved semiclassical probabilities become sensitive to the correct height of the barrier and deviate on the low or high side of the MHHJW probability. As mentioned above, the MHHJW probability can be derived from a symmetric Eckart potential. All the potentials that are the basis of the probabilities in Figure 6 are asymmetric. Had they been symmetric, then the case for $V_1 = D = 12$ kcal/mol would have produced an improved semiclassical probability identical with the MHHJW one. In Figure 6, the probability for the asymmetric potential is higher because a deeper product asymptote shaves away the product side of the barrier making it thinner.

In Figure 6, none of the probabilities go exactly to zero at the base of the barrier. The form of eq 1 would not allow this to be true unless $\theta(E)$ becomes infinitely large as E approaches zero. The barrier penetration integral cannot become infinitely large in this limit for any physically meaningful potential. However, due to high-order quantum mechanical effects not included in semiclassical theory, quantum mechanical probabilities do go to zero in this limit. It is instructive to see how they do so for the exact quantum dynamics tunneling probability P_{QD} for the pure Eckart potential.¹⁵ As developed in the Supporting Information (see eqs S4.1–S4.5), in the typical limit that $V_1 \gg |\hbar\omega_F|$, P_{QD} assumes the form:

$$\begin{aligned} P_{QD}(E) \sim & (1 - e^{\{4\pi\rho/[\hbar\omega_F(1+\rho)]\}\sqrt{V_1E}}) \\ & (1 - e^{\{4\pi\rho/[\hbar\omega_F(1+\rho)]\}\sqrt{V_1(E-V_1[1-\rho^2])}})P_{SC}(E) \end{aligned} \quad (28)$$

where $P_{SC}(E)$ is the semiclassical tunneling probability of eq 1 evaluated with the semiclassical $\theta(E)$ of eq 25 where $D_b = V_1$. Equation 28 in effect replaces the unit numerator of eq 1 with the product of two energy-dependent factors. The first factor goes to zero when $E = 0$ while the second factor goes to zero when $E = V_1 - V_2$, i.e., the zero of energy from the product side of the potential. A similar relationship between P_{QD} and $P_{SC}(E)$ exists for the inverted Morse oscillator with an exact quantum tunneling probability of the form:¹⁶

$$P_{QD}(E) = (1 - e^{-(8\pi/\hbar\omega_F)\sqrt{V_1E}})P_{SC}(E) \quad (29)$$

where $P_{\text{SC}}(E)$ is the semiclassical tunneling probability of eq 1 evaluated with the semiclassical $\theta(E)$ of eq 4. The multiplicative factor in the equation goes to zero when E goes to zero.

These comparisons with P_{QD} suggest a further but minor improvement in $P_{\text{SC}}(E)$ evaluated with the $\theta(E)$ of eqs 22–24 is its multiplication of the two threshold factors in eq 28. In Figure 7, the resulting probabilities with and without this threshold

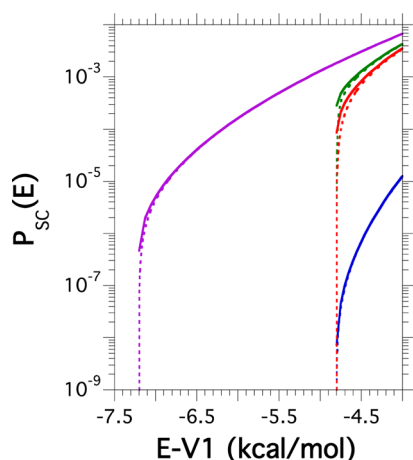


Figure 7. $P(E)$ with (dotted line) and without (solid line) threshold factors versus $E - V_1$ (see text for details). The color-coding is according to $(V_1$ [kcal/mol], ρ , $|\hbar\omega_F|$ [cm⁻¹]): red (4.8, 1.0, 2000), green (4.8, 1.1, 2000), blue (4.8, 1.0, 1000), and magenta (7.2, 1.0, 2000).

factor correction are displayed for four different potentials. All four potentials have $D = 12$ kcal/mol but vary in ρ , $|\hbar\omega_F|$ and V_1 . In every case, the corrections are localized to energies within a fraction of a kilocalorie per mole from the threshold. The largest correction is for the largest value of $|\hbar\omega_F|$ and the smallest values of ρ and V_1 . Such dependencies are as expected from eq 28.

4. DISCUSSION

In the previous section, the tunneling probability was developed in terms of a one-dimensional model. As stated at the start of that section, the multidimensional aspect of the original MHHJW formula can be recovered by interpreting E as the total energy minus the energy in the bound degrees of freedom and by $|\hbar\omega_F| \rightarrow |\hbar\Omega_F|$. The implications of a change in frequency merit some discussion. $|\hbar\Omega_F|$ through eq 6 is a function of all the quantum numbers of $F - 1$ degrees of freedom. The composite potential is a function of $|\hbar\Omega_F|$ and hence the action and corresponding tunneling probabilities all change with the change of any one quantum number among the bound degrees of freedom. The value of $|\hbar\Omega_F|$ influences the tunneling probabilities in two ways. First, given a composite potential, the action is inversely proportional to $|\hbar\Omega_F|$ as shown in the Supporting Information (see eq S3.5). The MHHJW formula has an identical dependence. This ultimately stems from the fact that the values of α used in the composite potential are through eq 16 all linear in $|\hbar\Omega_F|$. Second, D is dependent on the square of $|\hbar\Omega_F|$ through eq 7 and through eq 18 changes D_b which in turn influences the details of the matching between each outer segment of the composite potential and the middle segment. This produces a nonlinear dependence of the action on $|\hbar\Omega_F|$. The MHHJW formula does not have this dependence. Because of the analytic nature of the action, processing changes produced by $|\hbar\Omega_F|$ do not introduce important implementation problems. Most likely the dependence of the tunneling probability on bound state

quantum numbers is slight. Typically $|\hbar\omega_F| \gg |\hbar\omega_{FF}|$ and typically $|\hbar\omega_{FF}| \gg |\hbar\omega_F|$. From eq 6, this makes $|\hbar\Omega_F|$ typically only slightly dependent on bound state quantum numbers.

Although the focus of this article is on tunneling processes, the bound degrees of freedom perpendicular to the reaction path suffer from related limitations discussed for tunneling in Section 2. In particular, in analogy with eq 6 for the tunneling coordinate, eq 5 can be rearranged for the bound degrees of freedom as:

$$\Delta E = V_1 + \sum_{k=1}^{F-1} \{ \hbar\Omega_k(n_k + 1/2) + x_{kk}(n_k + 1/2)^2 \} - E \quad (30)$$

where

$$\hbar\Omega_k = \hbar\omega_k + \sum_{l=k+1}^{F-1} x_{kl}(n_l + 1/2) \quad (31)$$

If x_{kk} is negative, this is a Morse oscillator expression whose dissociation energy is the D of eq 7 with subscript $F \rightarrow k$. However, D is entirely composed of spectroscopic information at the bottom of the well that rests at the saddle point. The true dissociation energy is the change in energy from the saddle point caused by expanding the k th normal mode until a bond breaks and two asymptotic fragments form. This dissociation energy will equal D only if the bond fission exactly follows a Morse oscillator description. This is unlikely to be true even if a normal mode was an appropriate dissociation coordinate and certainly will be qualitatively untrue for many bending degrees of freedom where hindered rotations rather than bond fission are the rule. Consequently counting quantum numbers for either sums or densities of states cannot include progressions in eq 30 that exceed the value of n_k for the maximum energy D .

For typical applications of VPT2 to saddle points, the true dissociation energy of stretch modes or the hindered rotational barrier of bending modes are not calculated. If they were to be calculated, the same strategy used above for tunneling could be applied to incorporate this information. A native or composite potential could be constructed whose appropriate integration produces an action that reproduces both the VPT2 information of $\hbar\Omega_k$ and x_{kk} as well as the additional energetics information. Most likely, as in tunneling, the final action could be analytic. Semiclassical versions of sums and densities of states could then replace versions developed from state counting artificially limited by qualitatively incorrect dissociation energies.

VPT2 is the basis of the MHHJW formula and the ultimate cause of its limitations. However, it is well to note that a general semiclassical transition state theory can be formulated based on the full potential energy surface and not its local characterization by VPT2. Seideman and Miller¹⁸ as well as Hernandez and Miller¹⁹ show how semiclassical energies of the activated complex at the saddle point correspond to Siegert eigenvalues of the complex Hamiltonian matrix as expanded in harmonic oscillator functions where the one imaginary frequency leads to outgoing wave basis functions. These eigenvalues are a function of the quantum numbers n for the bound degrees of freedom and θ and therefore can be approximately represented by a least-squares fit to a basis set of functions of n and θ . Since these eigenvalues are equal to the energy E , this representation generalizes the VPT2 form of the right-hand side of eq 2 although undoubtedly with a more complicated form. The generalized equation can be approximately inverted to give θ as a numerical function of E and n , in analogy to eq 4. Since this process can

incorporate all relevant regions of the potential energy surface, it is qualitatively superior to the MHHJW formula but it achieves its superiority only at the price of qualitatively more *ab initio* electronics structure and semiclassical dynamics calculations. Such calculations are rarely, if ever, done. Relative to this general semiclassical theory, the theory described in this article is a rather simple approximation but one that is an improvement over the commonly used MHHJW formula.

5. SUMMARY

While correct for tunneling energies near the top of the barrier, we have shown that the analytic multidimensional semiclassical tunneling formula of Miller et al.⁵ is *qualitatively* incorrect for deep tunneling at energies well below the top of the barrier. The origin of this deficiency is that the height of the barrier in either the forward or reverse direction of the reaction is not directly incorporated into the formula. Instead, the formula uses an effective barrier that is only weakly related to the energetics but does correctly reproduce the harmonic description and anharmonic corrections of the reaction path as determined by second-order vibrational perturbation theory. The correct inclusion of the harmonic and anharmonic information was the goal of the Miller et al. semiclassical model and the reason for its reliability at energies near the top of the barrier.

Since the barrier is always known in the application of the formula, we have presented an improved semiclassical formula that correctly includes energetic information and allows a qualitatively correct representation of deep tunneling. This is done by constructing a three-segment composite Eckart potential that is continuous everywhere in both value and derivative. The middle segment of the composite reproduces to second order the original formula because it correctly reproduces both the harmonic and anharmonic descriptions of the reaction path at the saddle point. The first and third segments correctly reproduce the forward and reverse barrier, allowing the correct incorporation of the reaction energetics in both directions. This composite potential has an analytic barrier penetration integral from which the semiclassical action can be derived and then used to define the semiclassical tunneling probability. Although the resulting formula for an improved semiclassical reaction probability is more complicated than the original simple formula, it is stable and well behaved. The middle segment of the potential by itself produces an action formula as simple as the original formula but superior because it incorporates the asymmetry of the reaction barrier produced by the known reaction exoergicity. Comparison of the semiclassical and exact quantum tunneling probability for the Eckart potential suggests a simple threshold multiplicative factor to the improved formula to account for quantum effects very near threshold not represented by semiclassical theory.

The composite potential that is the basis of the improved semiclassical probability is not unique and cannot be made unique without further *ab initio* information. However, it does incorporate forward and reverse barrier heights and second-order vibrational perturbation theory information, it reproduces the original tunneling probabilities to second order, and its action and corresponding probabilities are all analytic. Several plausible alternatives both for a potential and for a direct derivation of the semiclassical action without reference to a potential are described along with their disadvantages and limitations.

For semiclassical descriptions based on second-order vibrational perturbation theory, the fundamental limitations for deep tunneling are also present for high-energy vibrations in normal

modes perpendicular to the reaction path at the saddle point. While harmonic and anharmonic properties of these vibrational modes are reliably described at energies near the saddle point, they are likely to be qualitatively incorrectly described at higher energies near hinder rotor barriers or bond dissociations. While the approach in this paper can be used to correct this problem, typically energetic information required for the correction is not available.

The original motivation of Miller et al. is more true now than when their paper was first published, namely that relatively scalable second-order vibrational perturbation theory calculations at saddle points can routinely provide anharmonic information much more cheaply than mapping out significant portions of the reaction path. The formulas in this paper will broaden the value of such calculations for applications in dynamics and kinetics.

The Supporting Information provides a largely self-contained derivation of all the formulas in this paper. Equally importantly, it contains a Fortran code, test input, and test output that implements the improved semiclassical tunneling formula. The code contains a driver and three major subroutines. One subroutine sets up the distance-independent constants of the potential and the consequent energy-independent constants of the action and probabilities. A second subroutine uses the constants to calculate the action and the probability at any input energy. A third optional subroutine uses the constants to calculate the potential at any input distance. The main driver routine loops over a range of input and calls the three subroutines. The test input will produce the test output that contains the potentials of Figure 3, the actions of Figure 4, and probabilities of Figure 6. Replacement of the main driver routine by user-specific calls to the subroutines will allow the implementation of the improved semiclassical probability developed in this paper.

■ ASSOCIATED CONTENT

■ Supporting Information

A largely self-contained derivation of all the formulas in this paper and Fortran code, test input, and test output that implements the improved semiclassical tunneling formula. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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Notes

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